

Sharp Error Amplification in Recursively Projected Multilinear Computations

Anonymous manuscript

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Abstract

We study finite-dimensional computations obtained by composing bounded multilinear maps along a rooted tree and applying an orthogonal projector after every internal operation. If each law has operator norm at most M and its normal output on already-projected inputs is at most ρ , then a tree with k internal vertices has projected root error at most $(k-1)\rho M^{k-1}$ times the product of the leaf norms. The missing root contribution is structural: the final projection removes the root's own normal defect. The coefficient is zero for $k = 1$ and the normalized constant is one for $k = 2$. For every ordered tree with exactly three internal vertices, arbitrary finite arities, independently selectable node laws, finite-dimensional real or complex Hilbert spaces, and orthogonal projectors, we determine the best dimension- and rank-independent constant:

$$W_3(\eta) = \begin{cases} \sqrt{4 - 3\eta^2}, & 0 < \eta \leq \sqrt{2/3}, \\ \frac{2}{\sqrt{3}\eta}, & \sqrt{2/3} \leq \eta \leq 1, \end{cases} \quad \eta = \rho/M.$$

The upper bound follows from orthogonal error coupling and a two-dimensional sum-of-squares identity; matching real two-dimensional constructions attain it. Thus local error accumulation is first-order sharp as $\eta \downarrow 0$, but the finite-defect extremal geometry is strictly smaller than the universal coefficient two. Exact constants at fixed η for $k \geq 4$ remain open. All novelty comparisons in this draft are provisional pending theorem-level bibliographic adjudication and independent review.

1 Introduction

Projection is a standard way to keep an evolving or recursively constructed object inside a prescribed reduced space. Its local effect is transparent: an orthogonal projector is contractive, and the discarded component can be measured. Its global effect is subtler. Once a discarded component enters a multilinear composition, subsequent maps can redirect it into the retained space. The final retained error is therefore neither the sum of local discarded norms nor the defect of the root alone.

This paper isolates the corresponding extremal problem. A finite ordered rooted tree specifies a composition of multilinear maps. Every internal value is projected before being passed upward. We assume a uniform operator-norm bound M and a uniform closure-defect bound ρ , measured only on inputs which are already in the appropriate projected ranges. The question is: *how large can the retained root error be, relative to the local budget ρ ?*

Our first result is the universal estimate

$$E_T^P \leq (k-1)\rho M^{k-1}L_T, \quad (1)$$

where k is the number of internal vertices and L_T is the product of leaf norms. The main result determines the exact normalized constant for $k=3$ as the piecewise function W_3 displayed in the abstract. The transition at $\eta_c = \sqrt{2/3}$ separates a budget-limited regime, in which an extremizer uses the full defect allowance, from a geometry-limited regime, in which the worst absolute error has already saturated.

Position relative to neighboring projection theories. Four nearby literatures make the scope easier to see.

- **HSVD and hierarchical tensor truncation.** Hierarchical SVD, hierarchical Tucker, and tensor-train truncation approximate a fixed tensor by a structured low-rank tensor and compare the truncation with best approximation in a tensor norm; representative quasi-optimality results are developed in [3, 4, 9]. Here the object is instead a composition of independently bounded multilinear maps, the denominator is a per-node closure budget, and the comparison is with the unprojected evaluation. Consequently the familiar $\sqrt{d}$ -type truncation factors and our $(k-1)/W_3(\eta)$ amplification constants answer different questions.
- **Alternating projections.** Classical alternating-projection theory studies products of projectors and convergence to an intersection; angles between subspaces control sharp error or rate bounds in important cases [2, 6]. Our projectors do not alternate on one state. They are interleaved with nonlinear multilinear laws on a tree, and the propagated component may re-enter a retained range. Orthogonality remains decisive, but through Pythagorean coupling at a node rather than through a Friedrichs angle between target subspaces.
- **Tensor-network perturbation and conditioning.** Sitewise perturbations of tensor networks already admit worst-case bounds, condition numbers, and tight witness constructions [10]; finite-precision analyses of tree tensor-network operations likewise propagate local roundoff and truncation errors through a contraction tree [1]. These are substantial precedents for sharp local-to-global amplification. Their perturbation variables are tensor sites or arithmetic operations, whereas our admissibility parameter controls the normal part of a node law only on already-projected inputs. They do not determine the fixed- η supremum for the projected-closure error in (2).
- **DLRA and TDVP.** Dynamical low-rank approximation and time-dependent variational principles project a vector field onto a tangent space of a fixed-rank manifold and analyze time-integration or variational error [5, 7, 8]. The present problem is static and finite, has no time step or tangent bundle, and allows a different multilinear law and projector at every vertex. It can serve as a local algebraic model of recursive projection, but it does not by itself establish a DLRA or TDVP error theorem.

Table 1: Theorem-level positioning from a scoped full-text audit. Failure to identify an equivalent theorem in that corpus is not a priority claim.

Result in this paper	Closest precedent	Difference requiring adjudication
Universal projected coefficient $k - 1$	layered/telescoping error bounds; Zhang–Solomonik sitewise perturbation; TTN finite-precision propagation	recursive laws with a projected-input closure budget and exact removal of the root defect; retained root error rather than contracted-tensor perturbation
$C_2^P(\eta) = 1$	equality in elementary operator-norm chains	exact class and closure convention may be unstated folklore
$C_3^{P,\text{fin}}(\eta) = W_3(\eta)$	Zhang–Solomonik tight worst-case site perturbation; exact alternating-projection constants	nonlinear fixed-defect- ratio supremum, piecewise phase transition, and attaining projected-law witness
$k = 3$ universality over arities and skeletons	tree-based tensor formats	classification by internal-child skeleton after freezing leaf slots

Table 1 is deliberately conservative. A scoped full-text audit did not identify an equivalent to the exact W_3 statement, but bibliographic novelty remains unestablished pending independent review. In particular, the paper does not claim to introduce sharp tensor-network stability or local-to-global amplification in general. Its mathematical center is the exact fixed- η projected-closure constant. The arity and skeleton universality below is a structural consequence of the two sharp three-node cases, not a separate broad novelty claim.

Contributions.

1. We prove the structural bound (1) directly by subtree induction and argument-wise telescoping.
2. We identify the exact constants at $k = 1, 2$ in the declared class.
3. For $k = 3$ we derive a contraction Gram lemma, reduce both possible tree skeletons to the same two-variable extremal problem, and solve it by a sum-of-squares identity.
4. We give explicit real two-dimensional witnesses and extend the result to every finite arity profile by freezing additional leaf slots.
5. We state the fixed- η problem for $k \geq 4$ as open, rather than extrapolating from the universal upper bound or from small cases.

2 Projected multilinear trees

Let T be a finite ordered rooted tree. Its internal vertices are $V_{\text{int}}(T)$, its leaves are $L(T)$, and $k = k(T) = |V_{\text{int}}(T)|$. Every vertex w carries a finite-dimensional Hilbert space H_w over $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$. An internal vertex v has ordered children $c_1(v), \dots, c_{m_v}(v)$, with $m_v \geq 2$, and a multilinear law

$$\mu_v : H_{c_1(v)} \times \cdots \times H_{c_{m_v}(v)} \longrightarrow H_v.$$

Every internal v also carries an orthogonal projector $P_v = P_v^* = P_v^2$. For a leaf ℓ set $\hat{P}_\ell = I_{H_\ell}$, and for an internal vertex set $\hat{P}_v = P_v$.

Given leaf data $z_\ell \in H_\ell$, define the ambient evaluation recursively by $F_\ell = z_\ell$ and

$$F_v = \mu_v(F_{c_1(v)}, \dots, F_{c_{m_v}(v)}).$$

The recursively projected evaluation is $R_\ell = z_\ell$ and

$$R_v = P_v \mu_v(R_{c_1(v)}, \dots, R_{c_{m_v}(v)}).$$

For the root r , the quantity of interest is

$$E_T^P = \|P_r F_r - R_r\|. \quad (2)$$

This differs from the ambient discrepancy $\|F_r - R_r\|$: it retains only the component visible in the reduced root space.

Definition 2.1 (Projected-input closure defect). For an internal vertex v , set

$$\rho_v^{\text{proj}} = \left\| (I - P_v) \mu_v(\hat{P}_{c_1(v)} \cdot, \dots, \hat{P}_{c_{m_v}(v)} \cdot) \right\|_{\text{op}}. \quad (3)$$

The restriction in (3) is load-bearing. It controls the normal output when the children have already been projected, exactly the inputs seen by the recursive evaluation. It is weaker than the unrestricted ambient defect $\|(I - P_v) \mu_v\|_{\text{op}}$, and the two conventions must not be interchanged.

A realization is (M, ρ) -admissible if $\|\mu_v\|_{\text{op}} \leq M$ and $\rho_v^{\text{proj}} \leq \rho$ at every internal vertex, where $M > 0$ and $0 < \rho \leq M$. Write

$$\eta = \rho/M \in (0, 1], \quad L_T = \prod_{\ell \in L(T)} \|z_\ell\|.$$

For a fixed tree T , define the dimension- and rank-independent constant over independently selectable node laws by

$$C_T^{P, \text{fin}}(\eta) = \sup \frac{E_T^P}{\rho M^{k-1} L_T}, \quad (4)$$

where the supremum ranges over all nondegenerate admissible finite-dimensional realizations with $\rho/M = \eta$. Equality in (4) is a class statement; it does not say that every fixed dimension and projector rank attains the supremum.

Scaling all node laws by a common positive factor and each leaf independently shows that the ratio in (4) is invariant. We may therefore normalize $M = 1$ and all nonzero leaves to unit norm. A zero leaf makes the entire multilinear evaluation vanish.

3 The universal projected bound

For a vertex v , let T_v be its subtree, let k_v be its number of internal vertices, and let L_v be the product of its leaf norms.

Lemma 3.1 (Subtree stability). *For every vertex v ,*

$$\|F_v\| \leq M^{k_v} L_v, \quad \|R_v\| \leq M^{k_v} L_v.$$

Proof. The statement is immediate at a leaf. At an internal vertex it follows by multilinearity, the operator-norm bound on μ_v , contractivity of P_v , and induction over the child subtrees. $\square$

Lemma 3.2 (Ambient subtree discrepancy). *For every vertex v ,*

$$\|F_v - R_v\| \leq k_v \rho M^{k_v-1} L_v.$$

Proof. The result is trivial at a leaf. Let v be internal and abbreviate the child values by F_i, R_i . Add and subtract $\mu_v(R_1, \dots, R_{m_v})$:

$$F_v - R_v = [\mu_v(F_1, \dots, F_{m_v}) - \mu_v(R_1, \dots, R_{m_v})] + (I - P_v)\mu_v(R_1, \dots, R_{m_v}).$$

The last term is bounded by $\rho M^{k_v-1} L_v$ because every R_i is in the corresponding projected range. Telescoping the first bracket one argument at a time gives one term for each child. Lemma 3.1 and the inductive hypothesis bound the i th term by $k_{c_i} \rho M^{k_v-1} L_v$. Since $k_v = 1 + \sum_i k_{c_i}$, summing proves the claim. $\square$

Theorem 3.3 (Structural $k-1$ bound). *Every admissible tree with $k \geq 1$ internal vertices satisfies*

$$\boxed{E_T^P \leq (k-1) \rho M^{k-1} L_T.} \quad (5)$$

Consequently $C_T^{P, \text{fin}}(\eta) \leq k-1$.

Proof. At the root,

$$P_r F_r - R_r = P_r [\mu_r(F_1, \dots, F_{m_r}) - \mu_r(R_1, \dots, R_{m_r})].$$

The root's local normal defect is absent because $P_r(I - P_r) = 0$. Telescoping over the root arguments and applying Lemma 3.2 to each child yields

$$E_T^P \leq \left(\sum_i k_{c_i} \right) \rho M^{k-1} L_T = (k-1) \rho M^{k-1} L_T.$$

$\square$

The improvement from k to $k-1$ is therefore not a cancellation inferred from examples. It follows from which component the final projector removes.

4 The cases $k=1$ and $k=2$

Corollary 4.1. *For $k=1$, $C_1^{P, \text{fin}}(\eta) = 0$.*

Proof. There is only the root, and by definition $R_r = P_r F_r$. $\square$

For $k=2$, after freezing any additional unit leaf slots, the internal-child skeleton is a chain. In the binary notation, let the first law act on leaves a, b , let the second law act on the first value and a leaf d , and put

$$D = (I - P_1)\mu_1(a, b), \quad R_1 = P_1\mu_1(a, b).$$

Then

$$E_T^P = \|P_2\mu_2(D, d)\|. \quad (6)$$

Theorem 4.2 (Exact two-vertex constant). *For the unrestricted finite-dimensional independent-law class,*

$$\boxed{C_2^{P,\text{fin}}(\eta) = 1 \quad (0 < \eta \leq 1).}$$

In a fixed binary realization, equality in the universal estimate holds if and only if

$$\begin{aligned} \|D\| &= \rho \|a\| \|b\|, \\ \|\mu_2(D, d)\| &= M \|D\| \|d\|, \\ \mu_2(D, d) &\in \text{Ran}(P_2). \end{aligned}$$

Proof. Equation (6) gives the upper bound and the three equality conditions by inspecting the chain

$$E_T^P \leq \|\mu_2(D, d)\| \leq M \|D\| \|d\| \leq \rho M \|a\| \|b\| \|d\|.$$

For attainment, normalize $M = 1$ and take $H = \mathbb{R}^2$ with basis e_0, e_1 and P the projection onto $\mathbb{R}e_0$. With unit leaves e_0 , choose

$$\mu_1(x, y) = \langle x, e_0 \rangle \langle y, e_0 \rangle (\sqrt{1 - \eta^2} e_0 + \eta e_1), \quad \mu_2(x, y) = \langle x, e_1 \rangle \langle y, e_0 \rangle e_0.$$

Both laws have norm at most one, the first closure defect is η , the root law has zero normal output, and $E_T^P = \eta$. Restoring scale gives $E_T^P = \rho M L_T$. Unit gates embed the same witness into additional leaf slots. $\square$

5 Three vertices: orthogonal coupling

The universal coefficient at $k = 3$ is two. It is not the exact finite-defect constant. Orthogonality couples the size of a discarded component to the size of the retained value from which it came.

At the first non-root internal vertex, write

$$D_1 = (I - P_1)F_1, \quad R_1 = P_1F_1.$$

Then $D_1 \perp R_1$ and, in the normalized setting,

$$\|D_1\|^2 + \|R_1\|^2 = \|F_1\|^2 \leq 1. \tag{7}$$

The second ingredient controls the cross term created when a contraction acts on two orthogonal directions.

Lemma 5.1 (Projector-contraction Gram bound). *Let $N : H \rightarrow K$ be linear with $\|N\|_{\text{op}} \leq 1$, let $u, v \in H$ be orthonormal, and let Q be an orthogonal projector on K . If $s = \|QNv\|$, then*

$$|\langle Nu, QNv \rangle| \leq s\sqrt{1 - s^2}. \tag{8}$$

Proof. Set $A = N^*QN$. Then $0 \preceq A \preceq I$. On $\text{span}\{u, v\}$ write $A_{vv} = s^2$, $A_{uu} = \alpha$, and $A_{uv} = c$. Positivity of A gives $|c|^2 \leq \alpha s^2$, while positivity of $I - A$ gives $|c|^2 \leq (1 - \alpha)(1 - s^2)$. The largest possible minimum of these two quantities occurs at $\alpha = 1 - s^2$, proving $|c| \leq s\sqrt{1 - s^2}$. $\square$

5.1 The chain skeleton

Normalize $M = L_T = 1$ and $\rho = \eta$. Put $q = \|D_1\| \leq \eta$ and $p = \|R_1\| \leq \sqrt{1 - q^2}$. Freezing the leaf inputs at the next vertex reduces its law to a linear contraction N . With $Q = I - P_2$ and suitable orthonormal u, v , put $s = \|QNv\| \leq \eta$. The discrepancy that can reach the projected root has the form $qNu + pQNv$. Lemma 5.1 therefore gives

$$\|qNu + pQNv\|^2 \leq q^2 + p^2s^2 + 2qps\sqrt{1 - s^2}. \quad (9)$$

The right-hand side increases with p , so take $p = \sqrt{1 - q^2}$. Write $q = \sin \alpha$, $s = \sin \beta$ and $\xi = \tan \alpha$, $\zeta = \tan \beta$. The squared objective becomes

$$f(\xi, \zeta) = \frac{(\xi + \zeta)^2 + \xi^2\zeta^2}{(1 + \xi^2)(1 + \zeta^2)}. \quad (10)$$

The extremal calculation is encoded in the identity

$$4(1 + \xi^2)(1 + \zeta^2) - 3[(\xi + \zeta)^2 + \xi^2\zeta^2] = (\xi\zeta - 2)^2 + (\xi - \zeta)^2. \quad (11)$$

Hence $f \leq 4/3$, with equality exactly at $\xi = \zeta = \sqrt{2}$, or $q = s = \sqrt{2/3}$. Under the box constraints $q, s \leq \eta$, the maximum is at $q = s = \eta$ until that point becomes available and remains $4/3$ thereafter. Dividing the maximal absolute error by η gives

$$W_3(\eta) = \begin{cases} \sqrt{4 - 3\eta^2}, & 0 < \eta \leq \sqrt{2/3}, \\ \frac{2}{\sqrt{3}\eta}, & \sqrt{2/3} \leq \eta \leq 1. \end{cases} \quad (12)$$

Theorem 5.2 (Exact chain constant). *For the ordered three-vertex chain with independently selectable laws,*

$$C_{3,\text{chain}}^{P,\text{fin}}(\eta) = W_3(\eta).$$

Proof. The preceding argument gives the upper bound. For attainment take $H = \mathbb{R}^2$, $P = e_0e_0^*$, all leaves equal to e_0 , and

$$t = \min\{\eta, \sqrt{2/3}\}, \quad p = c = \sqrt{1 - t^2}.$$

Define

$$\begin{aligned} \mu_1(x, y) &= \langle x, e_0 \rangle \langle y, e_0 \rangle (te_1 + pe_0), \\ \mu_2(x, y) &= \langle y, e_0 \rangle N_t x, \end{aligned}$$

where the orthogonal map N_t satisfies

$$N_t e_0 = te_1 - ce_0, \quad N_t e_1 = ce_1 + te_0.$$

After two levels the propagated discrepancy is

$$v = tN_t e_1 + pt, e_1 = t^2 e_0 + 2tp, e_1, \quad \|v\| = t\sqrt{4 - 3t^2}.$$

Let $Ah = \langle v, h \rangle \|v\|^{-1} e_0$ and $\mu_3(x, y) = \langle y, e_0 \rangle Ax$. All three laws are contractions, both active defects equal $t \leq \eta$, and the root defect is zero. The projected root error is $\|v\|$, which gives (12) after division by η . $\square$

Remark 5.3 (Two regimes). The absolute normalized extremal error is

$$G_3(\eta) := \eta W_3(\eta) = \begin{cases} \eta \sqrt{4 - 3\eta^2}, & \eta \leq \sqrt{2/3}, \\ 2/\sqrt{3}, & \eta \geq \sqrt{2/3}. \end{cases}$$

Below the transition the witness uses the full defect budget. Above it, the effective leakage freezes at $\sqrt{2/3}$; the decrease of W_3 results from normalizing a saturated absolute error by a growing allowed budget.

5.2 The branching skeleton

Now two internal children feed the root. Write their discarded and retained norms as q_i and p_i , with $q_i \leq \eta$ and $p_i \leq \sqrt{1 - q_i^2}$. After scalarizing the projected root output, maximizing over contractive bilinear forms is dual to a nuclear norm. The relevant two-by-two coefficient matrix is

$$A = \begin{pmatrix} q_1 q_2 & q_1 p_2 \\ p_1 q_2 & 0 \end{pmatrix}.$$

Since $\|A\|_*^2 = \|A\|_F^2 + 2|\det A|$, saturation of the p_i bounds gives

$$\|A\|_*^2 = q_1^2 + q_2^2 - q_1^2 q_2^2 + 2q_1 q_2 \sqrt{1 - q_1^2} \sqrt{1 - q_2^2}. \quad (13)$$

With $q_1 = \sin \alpha$, $q_2 = \sin \beta$, expression (13) is exactly $f(\tan \alpha, \tan \beta)$ from (10). Thus the same sum-of-squares identity controls both skeletons.

Theorem 5.4 (Exact branching constant). *For the ordered three-vertex branching tree with independently selectable laws,*

$$C_{3,\text{branch}}^{P,\text{fin}}(\eta) = W_3(\eta).$$

Proof. Equation (13) and (11) give the upper bound. For attainment, use two copies of the first-node output $te_1 + pe_0$ from Theorem 5.2, with $t = \min\{\eta, \sqrt{2/3}\}$. At the root choose a norm-one bilinear scalar form given by a polar factor of the coefficient matrix A , and direct that scalar into $\text{Ran}(P_r)$. Operator/nuclear-norm duality then attains $\|A\|_* = t\sqrt{4 - 3t^2}$, while the root closure defect is zero. This is a real two-dimensional rank-one-projector realization and yields $W_3(\eta)$. $\square$

6 Structural universality at $k = 3$

Deleting the leaves from a rooted tree with exactly three internal vertices leaves only two possible internal-child skeletons: a chain, or two siblings feeding the root. Higher arity adds only leaf-occupied slots.

Fixing a unit leaf in such a slot converts a multilinear law into an effective law on its internal-child slots. This operation does not increase the operator norm. Because leaves carry the identity extended projector, it also does not increase the projected-input closure defect. The effective law is linear at a vertex with one internal child and bilinear at the root of the branching skeleton. Conversely, the two-dimensional witnesses above embed in every finite arity profile by inserting unit e_0 gates in the extra leaf slots.

Theorem 6.1 (Universal exact constant for three vertices). *Let T be any finite ordered rooted tree with exactly three internal vertices and internal arities at least two. Then every admissible finite-dimensional real or complex realization satisfies*

$$E_T^P \leq W_3(\eta) \rho M^2 L_T.$$

Moreover, over the unrestricted finite-dimensional class with independently selectable node laws,

$$\boxed{C_T^{P,\text{fin}}(\eta) = W_3(\eta) \quad (0 < \eta \leq 1).} \quad (14)$$

The equality is attained over $\mathbb{R}$ in dimension two with rank-one projectors.

Proof. Freeze unit leaf slots to reduce to the chain or branching analysis, giving the upper bound without increasing either admissibility budget. Embed the matching two-dimensional witness into the original arity profile with unit gates to obtain the lower bound. $\square$

Remark 6.2 (Quantifiers). Theorem 6.1 determines the best constant uniform in finite dimension, rank, arity, and skeleton. It does not assert equality in each fixed realization. For example, if every $P_v = I$, then the projected error is identically zero.

For $0 < \eta < \sqrt{2/3}$,

$$W_3(\eta) = 2 - \frac{3}{4}\eta^2 - \frac{9}{64}\eta^4 + O(\eta^6). \quad (15)$$

Thus the universal coefficient two is asymptotically correct to first order, while orthogonality forces a strict second-order improvement at every positive defect ratio.

7 What remains open

Theorem 3.3 applies to every finite tree, but an upper bound is not an exact constant. For a fixed tree with $k \geq 4$ internal vertices, the central unresolved finite-defect problem is

$$\boxed{C_T^{P,\text{fin}}(\eta) = ? \quad (0 < \eta \leq 1).}$$

The $k = 3$ proof does not extend by independent repetition: a larger tree couples several retained and discarded components, and the two-variable Gram geometry is replaced by a higher-dimensional compatibility problem. The following questions are therefore left open.

1. Determine $C_T^{P,\text{fin}}(\eta)$ at fixed η for $k = 4$, separately for the distinct internal-child skeletons.
2. Decide whether the leading finite-defect correction to $k - 1$ is always quadratic in η and whether its coefficient is a combinatorial invariant of T .
3. Determine how exact constants change under law sharing, fixed dimensions or projector ranks, and oblique rather than orthogonal projectors.

These are mathematical questions about the declared extremal class. Numerical search may suggest witnesses, but cannot establish a universal constant.

8 Limitations and novelty boundary

The results are finite-dimensional and deterministic. They require orthogonal projectors, independently selectable node laws for the unrestricted sharpness statements, and the projected-input closure convention (3). They do not imply continuum limits, time-integration error bounds or tensor-rank quasi-optimality.

The bibliography records a scoped full-text comparison with HSVD/HT, tensor-network perturbation, alternating projections, and DLRA/TDVP. No equivalent to (14) was identified in that corpus, but this negative search result is not a proof of novelty. The citation neighborhood of sharp tensor-network conditioning, the original Kayalar–Weinert paper, and the approximate-multiplicativity literature still require independent human review. The appropriate claim is therefore *we determine the exact constant in the declared model*, not a priority claim.

9 Conclusion

Recursive projection through a multilinear tree has a simple universal budget and a nontrivial exact geometry. The retained root error receives no local contribution from the root itself, which yields the coefficient $k - 1$. At three internal vertices, however, orthogonality prevents simultaneous saturation of the two propagated sources. Both possible skeletons reduce to the same sum-of-squares problem, giving the exact constant $W_3(\eta)$ and a phase transition at $\eta = \sqrt{2/3}$. Determining the corresponding fixed-defect constants for $k \geq 4$ is the next mathematical problem.

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